

Today's Agenda

Final project presentation & report requirements

Edge-based segmentation

Region-based segmentation

Announcement

We will have Quiz #2 on Tuesday, April 14.

Open book and open notes

Final Exam Schedule

Final exam has been scheduled:

12:30 p.m. – 3:00 p.m., Tuesday, May 5

Requirement:

- **Closed book and closed notes**
- **One-page double-sided cheat sheet is allowed**
- **A calculator is allowed for $+ - * /$**

It will cover all the topics discussed in class

Final-Project Presentation

- **Team project – a single presentation:** each team has **15** minutes (13 minutes for presentation and 2 minute for questions) to present your project.
 - Both members should take part in the presentation.
- **Single person project:** Each student has **13** minutes (11 minutes for presentation and 2 minute for questions) to present your project.

Final-Project Presentation

Presentation days:

- Thursday, April 16
- Tuesday, April 21
- Thursday, April 23

Send me an email (tongy@cse.sc.edu) by 11:59pm, April 5 including preference of these three days in a decreasing order. Earlier email has higher priority in choosing the day

What Should Be Included In the Final-Project Presentation

For a research/hands-on project

- An introduction of the background
- A brief literature review
- Methodology of your proposed method
- Experiment setup: dataset, evaluation metrics, etc.
- Experimental results if any
- Conclusion
- Reference list

Note: You don't need to have the experimental results by presentation date, but you need to include the information about the experimental setup and implementation details.

What Should Be Included In the Final-Project Presentation

For a survey project

- An introduction of the background
- An organized structure of the papers
- Comparison and critical comments of the methods/groups you reviewed
- Conclusion
- Reference list

Note: You don't need to complete the survey project by presentation date, but you need to show how you organize your papers. You can cover part of the papers in your full list.

Final Presentation Criteria

Category	Description	Worth
Description of Problem	Did you make clear the nature of the problem you were trying to solve?	15
Description of Methodology (Research project)	Did you present the methodology clearly? Did you provide sufficient (key) information to the audience?	25
Description of Experiment (Research project)	Could your experiment design – dataset, evaluation metrics, baseline approaches effectively demonstrate your proposed method?	25
Discussion and Comparison (Survey project)	Did your discussion clearly show your understanding of the methods? Did you make critical comments on the methods?	50
Visual Aids	Were visual aids used effectively? Were slides clear and easy to read by the audience?	10
Clarity & Organization	Was the presentation easy to understand; did it have a logical flow and organization?	15
Timing	Was the presentation well-paced? Did it fit within the time allotted?	10

On the Final Project Report

Written report due time: 11:59pm. May 5th

- Report format: the same as a conference paper
 - For example, you can use a template for ICIP 2025
https://cmsworkshops.com/ICIP2025/papers/paper_kit.php#FileFormat
 - You need to follow a formal citation format, e.g., IEEE format
- Length: 4-8 pages double-column (+ reference, no page limit on the reference)
- Code must be submitted with clear comments (Research project only)

Academic integrity (avoiding plagiarism)

- don't copy other person's work
- describe using your own words
- complete citation and acknowledgement whenever you use any other work (either published or online)

Generative Artificial Intelligence Policy

In this course, students shall give credit to AI tools whenever used, even if only to generate ideas rather than usable text or illustrations. When using AI tools, add an appendix showing

- the entire exchange, highlighting the most relevant sections;
- a description of precisely which AI tools were used (e.g. ChatGPT private subscription version or DALL-E free version),
- an explanation of how the AI tools were used (e.g. to generate ideas, turns of phrase, elements of text, long stretches of text, lines of argument, pieces of evidence, maps of the conceptual territory, illustrations of key concepts, etc.);
- an account of why AI tools were used (e.g. to save time, to surmount writer's block, to stimulate thinking, to handle mounting stress, to clarify prose, to translate text, to experiment for fun, etc.).

Use AI tools during quizzes, exams, or homework assignments is NOT allowed unless explicitly permitted and instructed.

Requirement for Final Project

In form of a complete research project

- Introduction (problem formulation/definition)
- A brief literature review
- The proposed method and analysis
- Experiment including experiment setup and experimental results
- Conclusion
- Reference

A survey research

- A well-defined problem or topic
- An organized structure of the reviewed papers
- Analyze/discuss these methods and compare them, preferred in groups
- Conclusion
- Reference

Note: you need to include ≥ 15 references

Final Project Report Grading Criteria

Category	Description	Worth
Description of Problem	Did you make clear the nature of the problem you were trying to solve?	10
Description of Methodology (Research project)	Did you present the methodology clearly? Did you provide sufficient (key) information to the readers?	35
Description of Experiment (Research project)	Could your experiments effectively demonstrate your proposed method?	30
Discussion and Comparison (Survey project)	Was the problem comprehensively reviewed? Did your discussion clearly show your understanding of the methods? Did you make critical comments on the methods you reviewed? Were the reviewed papers well organized? Did you show the connections among the papers?	65
Writing Clarity	Does the report read well? Is it easy to understand?	10
Organization & Length	Is the report well-organized? Does it have a logical flow?	10
Reference	Is the reference section complete and in consistent format; are the citations in the text in consistent format?	5

Canny Edge Detector

The general goal:

1. Low error rate: all edges should be found and there should be no false response
2. Edge points should be well localized: the edges located must be as close as possible to the true edges
3. Single edge point response: the detector should return only one point for each true edge point

Canny edge detector: expressing these three criteria mathematically and then find optimal solutions

Canny Edge Detector

Gaussian smoothing + 1st order derivative

→ optimal step edge detector

$$f_s(x, y) = G(x, y) \otimes f(x, y)$$

$$g_x = \partial f_s / \partial x \quad g_y = \partial f_s / \partial y$$

$$M(x, y) = \sqrt{g_x^2 + g_y^2} \quad \alpha(x, y) = \tan^{-1}(g_y / g_x)$$

Thick edge Nonmaxima suppression → Single edge

Quantize the Edge Direction

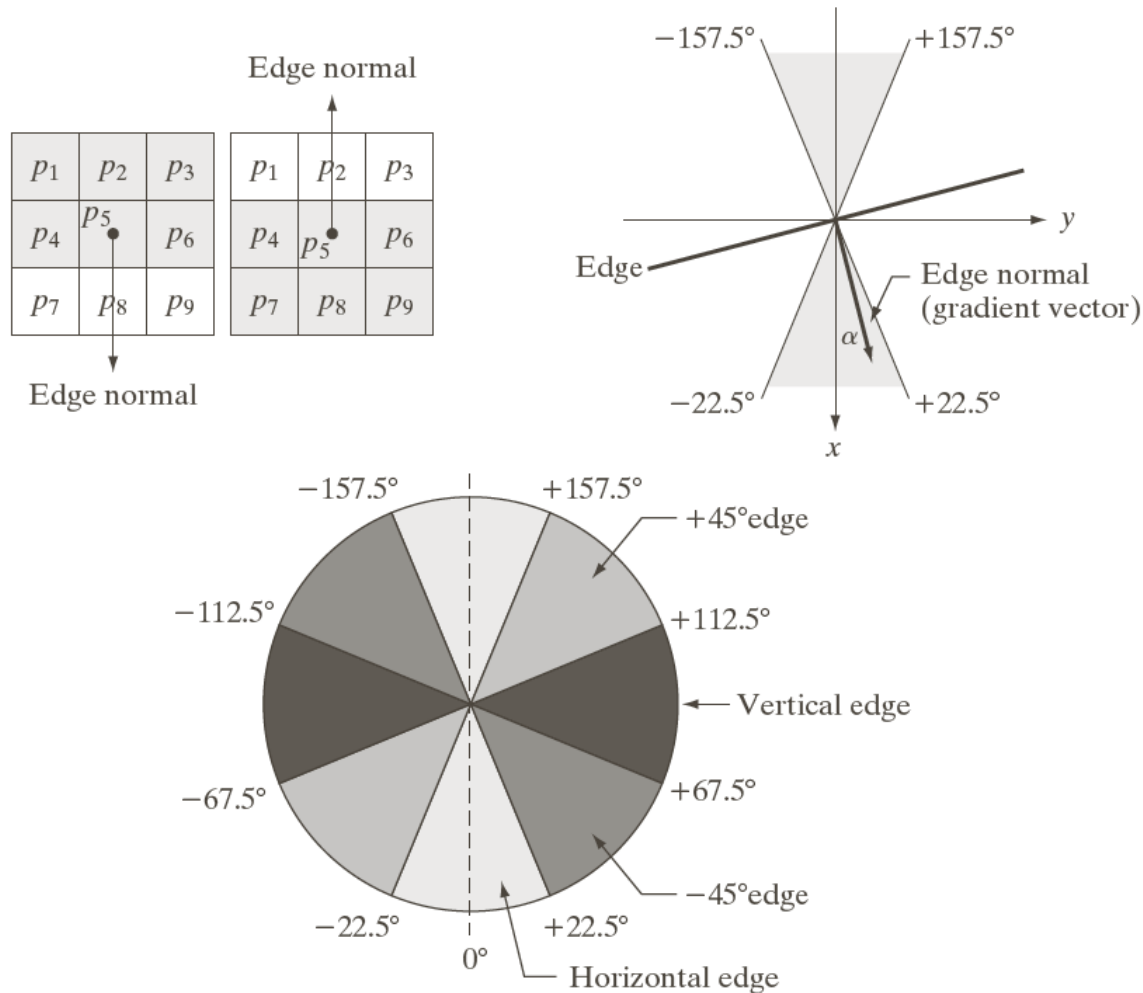
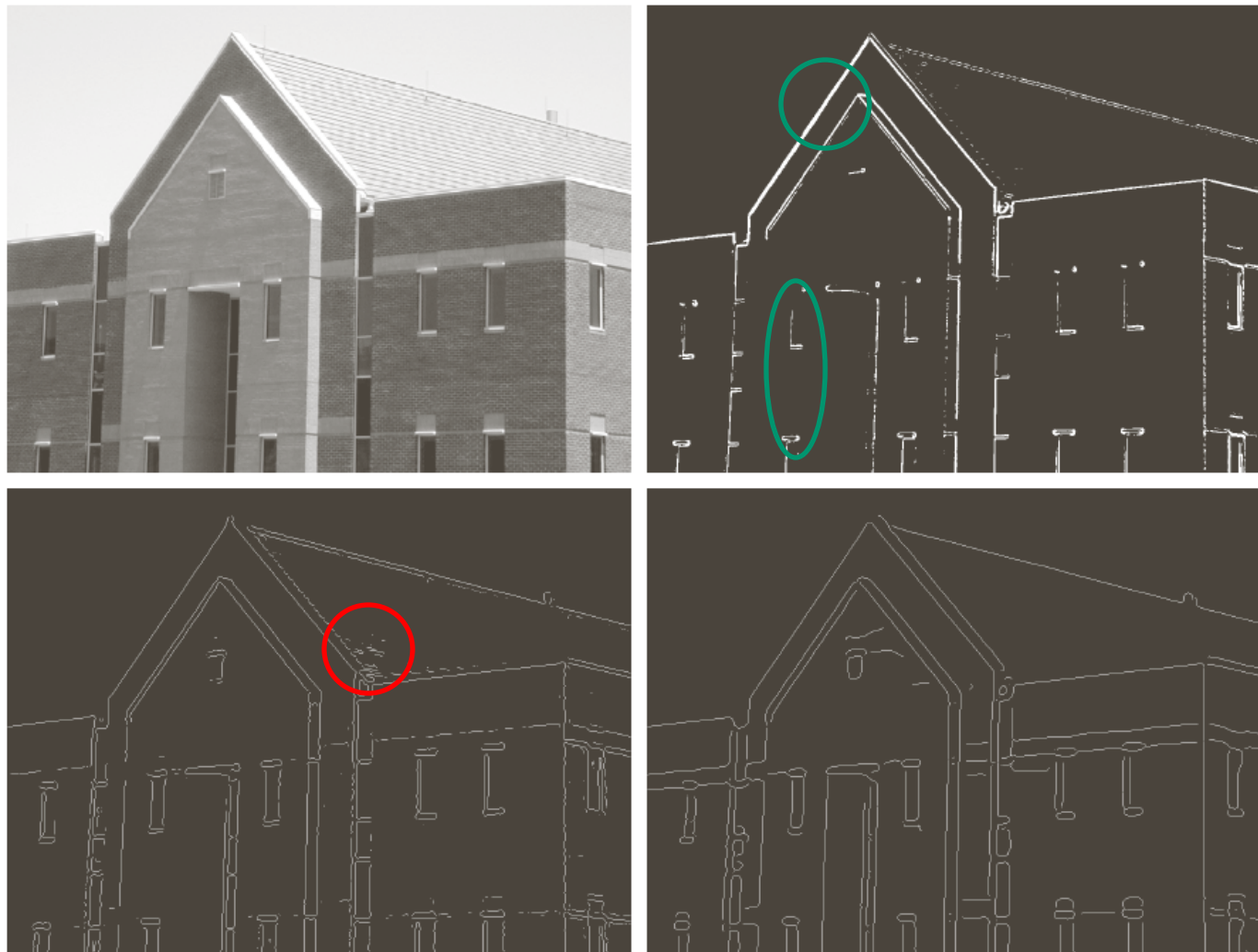


FIGURE 10.24
 (a) Two possible orientations of a horizontal edge (in gray) in a 3×3 neighborhood.
 (b) Range of values (in gray) of α , the direction angle of the edge normal, for a horizontal edge.
 (c) The angle ranges of the edge normals for the four types of edge directions in a 3×3 neighborhood. Each edge direction has two ranges, shown in corresponding shades of gray.

Canny Detector -- Algorithm

1. Smooth the input image with a Gaussian filter
2. Compute the gradient magnitude and angle images
3. Apply nonmaximum suppression on the gradient magnitude image
 1. At (x, y) , find the quantized edge normal d_k
 2. If the value $M(x, y)$ is less than at least one of its two neighbors along d_k , let $g_N(x, y) = 0$; otherwise $g_N(x, y) = M(x, y)$
4. Reduce false edge: double thresholding and connectivity analysis to detect and link edges
 1. High-threshold $\rightarrow$ strong edge pixels $\rightarrow$ valid edge pixels
 2. Low-threshold $\rightarrow$ weak edge pixels $\rightarrow$ valid only when connected to strong edge pixels

An Example



a	b
c	d

FIGURE 10.25

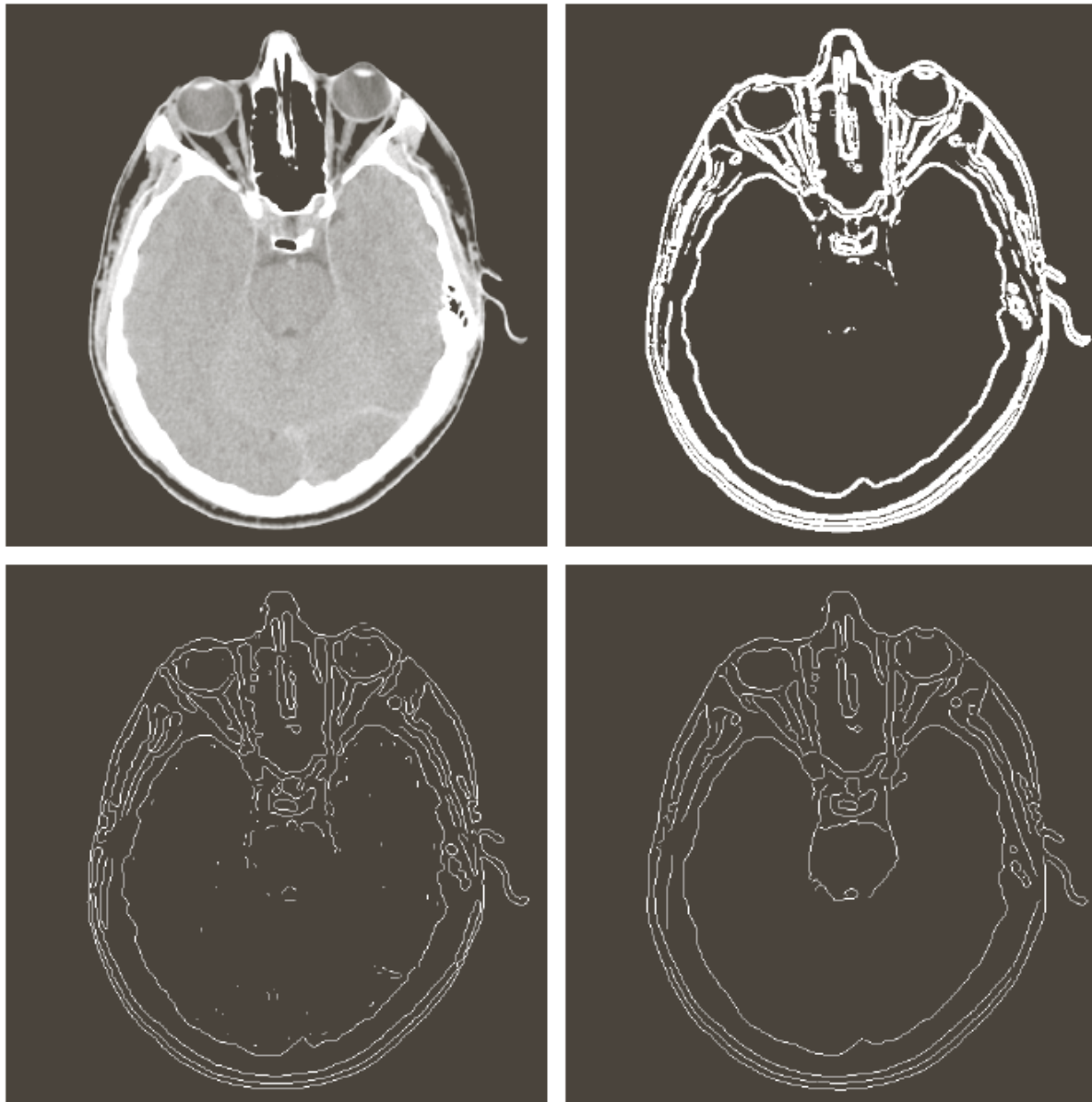
(a) Original image of size 834×1114 pixels, with intensity values scaled to the range $[0, 1]$.

(b) Thresholded gradient of smoothed image.

(c) Image obtained using the Marr-Hildreth algorithm.

(d) Image obtained using the Canny algorithm. Note the significant improvement of the Canny image compared to the other two.

An Example



a	b
c	d

FIGURE 10.26

(a) Original head CT image of size 512×512 pixels, with intensity values scaled to the range $[0, 1]$.
(b) Thresholded gradient of smoothed image.
(c) Image obtained using the Marr-Hildreth algorithm.
(d) Image obtained using the Canny algorithm.
(Original image courtesy of Dr. David R. Pickens, Vanderbilt University.)

Edge Linking and Boundary Detection

All edge detection algorithms can only detect fragments of boundaries, due to image noise, non-uniform illuminations, or other effects

Edge linking: link edges into longer meaningful edges or a full region boundaries

Three classic methods:

- Local processing
- Regional Processing
- Hough transform

Edge Linking – Local Processing

Link the edge points with similar properties:

- Strength
- Direction

Two edge pixels are linked if

$$|M(s, t) - M(x, y)| \leq E$$

$$|\alpha(s, t) - \alpha(x, y)| \leq A$$

A Local Processing Example

Objective: find rectangular region for license plate recognition.



a	b	c
d	e	f

FIGURE 10.27 (a) A 534×566 image of the rear of a vehicle. (b) Gradient magnitude image. (c) Horizontally connected edge pixels. (d) Vertically connected edge pixels. (e) The logical OR of the two preceding images. (f) Final result obtained using morphological thinning. (Original image courtesy of Perceptics Corporation.)

Global Processing

A basic idea: Given n points

1. Find $n(n - 1)/2$ lines between each pair of points
2. Find all subset of points that are close to particular lines. This needs $n(n - 1)n/2$ comparisons

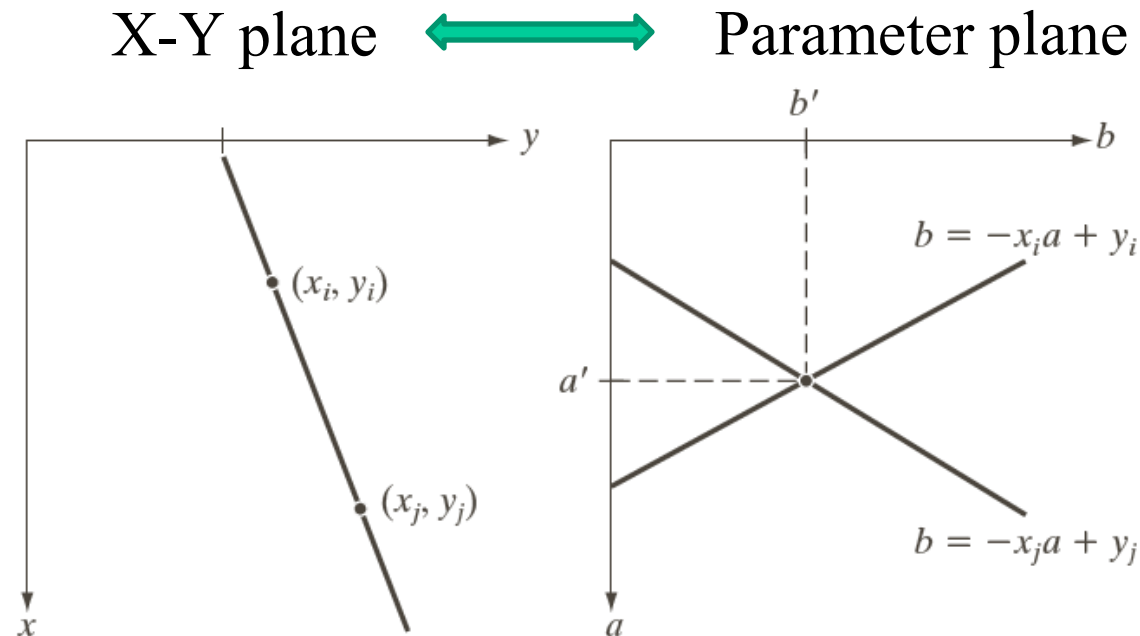
This is computationally expensive!



Hough transform

Hough Transform

A line in x-y plane is a point in the parameter plane.
A point in x-y plane is a line in the parameter plane.



a b

FIGURE 10.31
(a) xy-plane.
(b) Parameter space.

Problem of slope-intercept form: the slope a approaches infinity for vertical lines

Hough Transform

A line in x-y plane is a point in the parameter plane.
A point in x-y plane is a sinusoidal in the parameter plane (polar space).

$$x \cos \theta + y \sin \theta = \rho$$

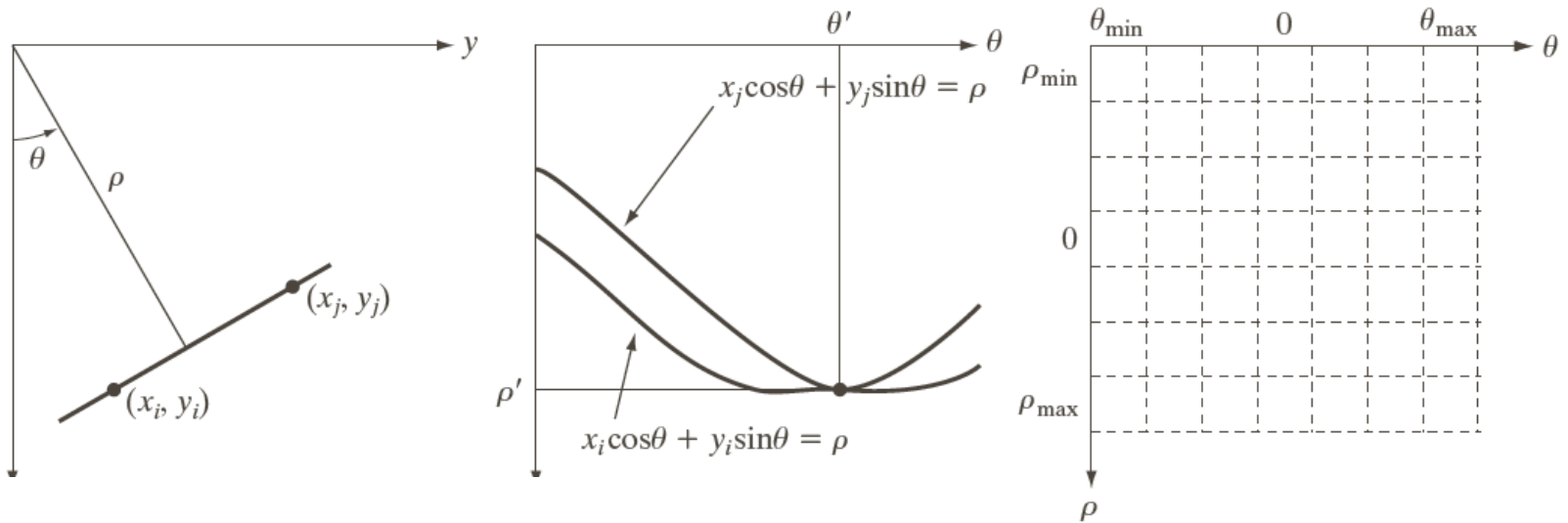
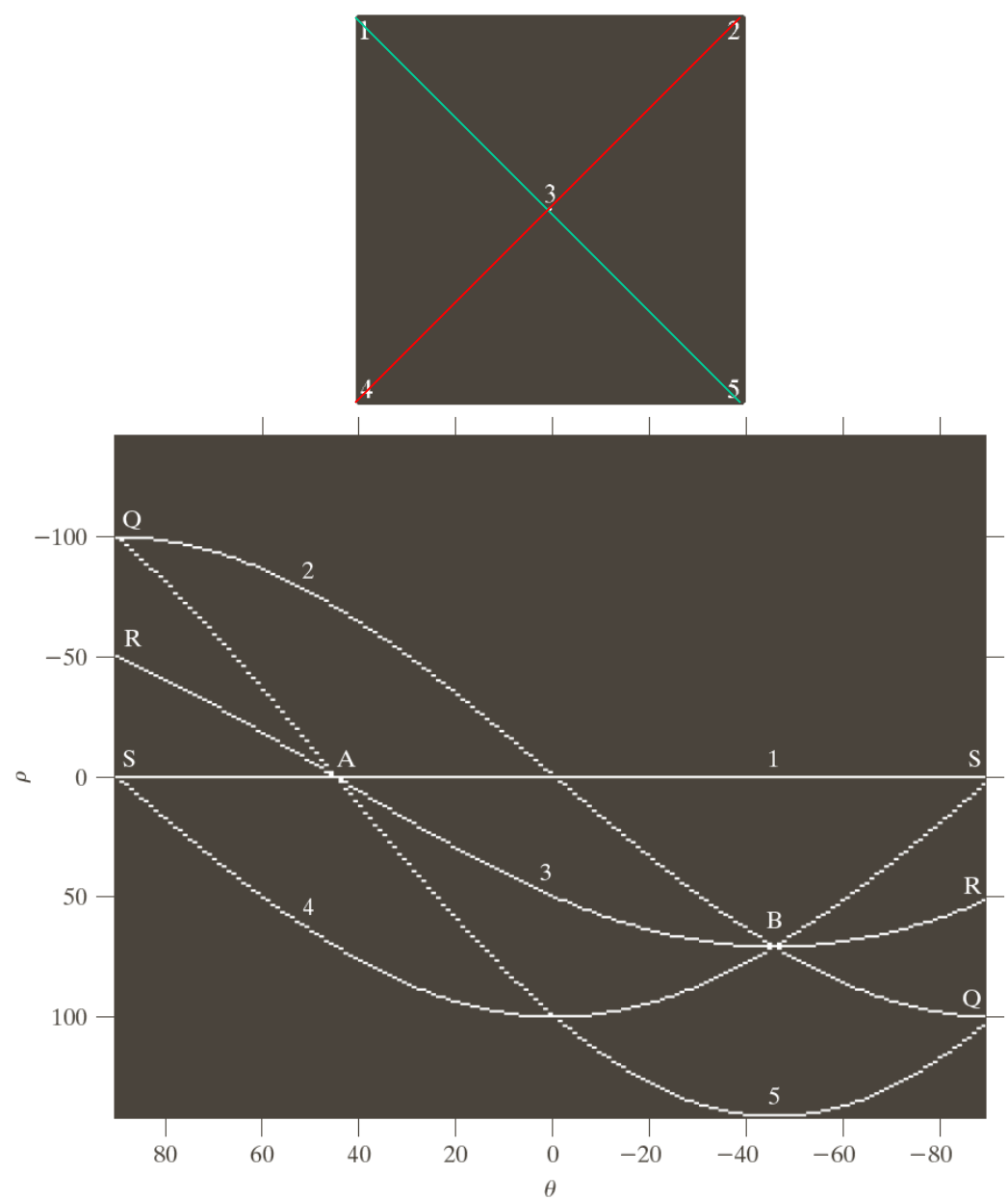


FIGURE 10.32 (a) (ρ, θ) parameterization of line in the xy -plane. (b) Sinusoidal curves in the $\rho\theta$ -plane; the point of intersection (ρ', θ') corresponds to the line passing through points (x_i, y_i) and (x_j, y_j) in the xy -plane. (c) Division of the $\rho\theta$ -plane into accumulator cells.

A Toy Example



a
b

FIGURE 10.33
(a) Image of size 101×101 pixels, containing five points.
(b) Corresponding parameter space.
(The points in (a) were enlarged to make them easier to see.)

A: green line
B: red line

Hough Transform – Real Example

Find the runway

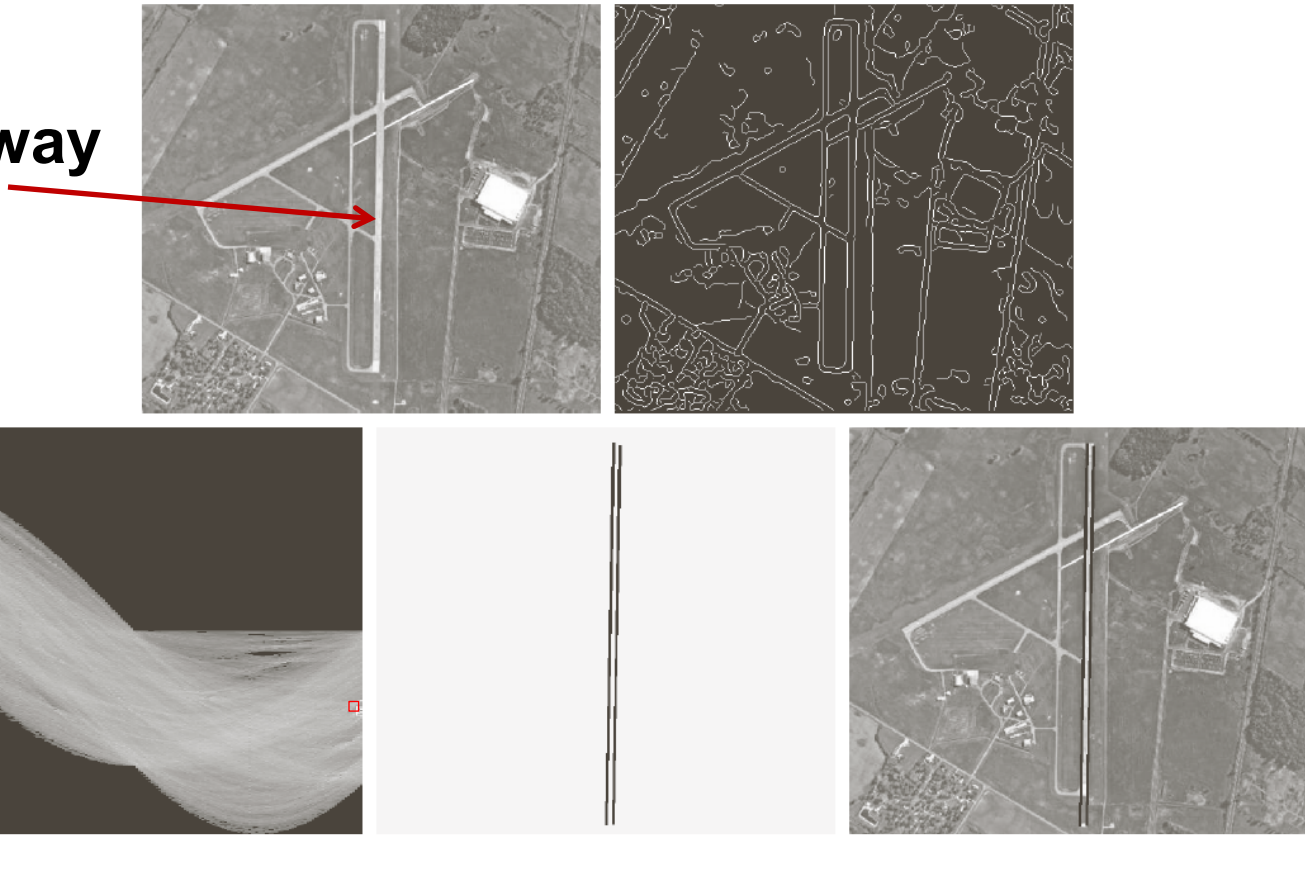
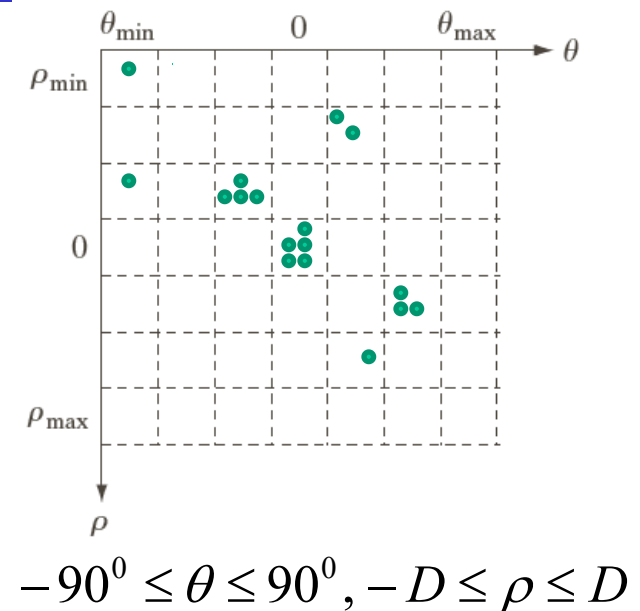


FIGURE 10.34 (a) A 502×564 aerial image of an airport. (b) Edge image obtained using Canny's algorithm. (c) Hough parameter space (the boxes highlight the points associated with long vertical lines). (d) Lines in the image plane corresponding to the points highlighted by the boxes. (e) Lines superimposed on the original image.

Hough Transform Algorithm

- Obtain a binary edge image using any edge detector
- Specify subdivisions in the ρ - θ plane
- For each edge point
 - For each θ value in the accumulator cell
 - Update the accumulator cell with the corresponding ρ
- Examine the counts of accumulator cell for high pixel concentrations
- For each chosen cell, link the pixels based on the continuity



Edge-based Segmentation

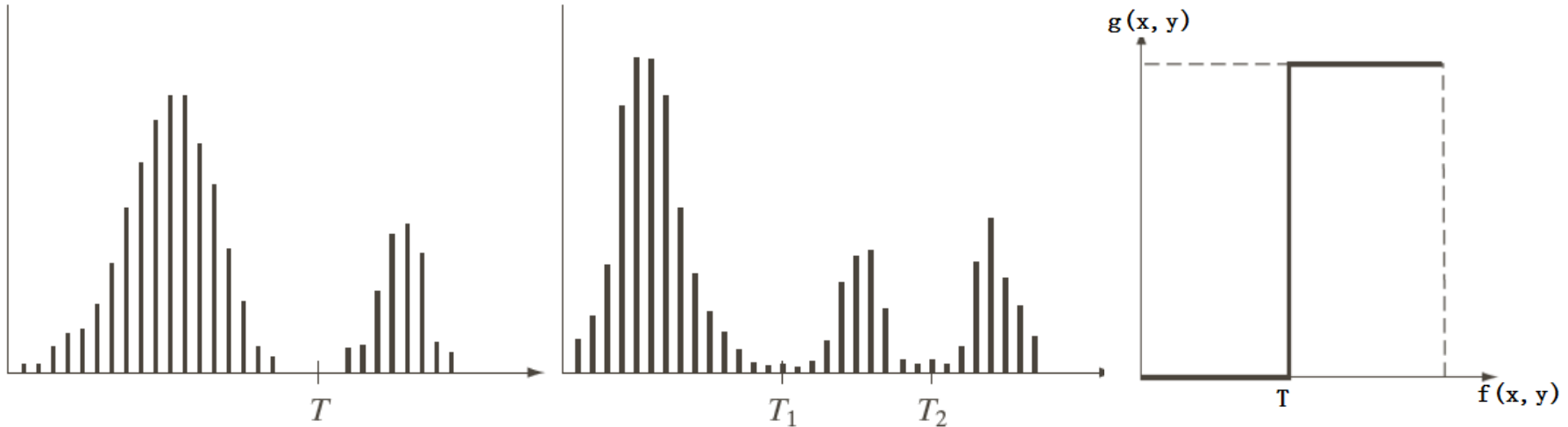
Thresholding is required to get edge pixels and realize edge-based segmentation

Boundary detection (edge linking) is still a hot research topic in image processing and computer vision

Incorporate domain knowledge:

- Psychology rules on the boundary: smooth, convex, symmetry, closed, complete, etc
- Template shape information: hand, stomach, lip, etc
- Appearance information: region-based texture, intensity/color, etc.

Edge-based Segmentation -- Intensity Thresholding



Object/background segmentation:

$$g(x, y) = \begin{cases} 1 & \text{if } f(x, y) > T \\ 0 & \text{if } f(x, y) \leq T \end{cases}$$

- A constant T - global thresholding
- A variable T - local/regional thresholding; adaptive thresholding
- Multiple T - multiple thresholding

Key Factors Affect Thresholding

- Separation between peaks
- Noise level
- Relative sizes of objects and background
- Uniformity of the illumination source
- Uniformity of the reflectance of the image

The Role of Noise in Image Thresholding

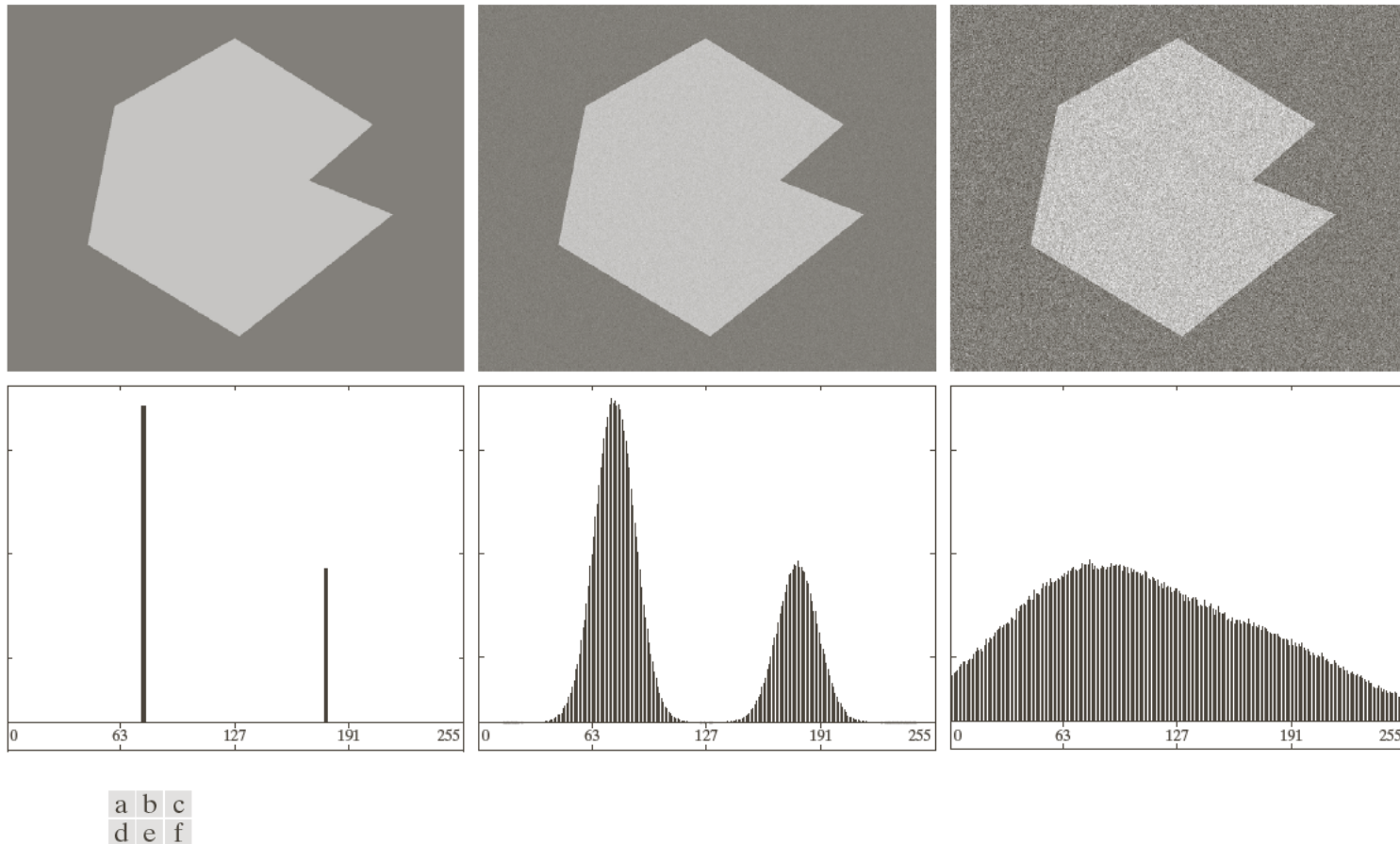


FIGURE 10.36 (a) Noiseless 8-bit image. (b) Image with additive Gaussian noise of mean 0 and standard deviation of 10 intensity levels. (c) Image with additive Gaussian noise of mean 0 and standard deviation of 50 intensity levels. (d)–(f) Corresponding histograms.

The Role of Illumination in Thresholding

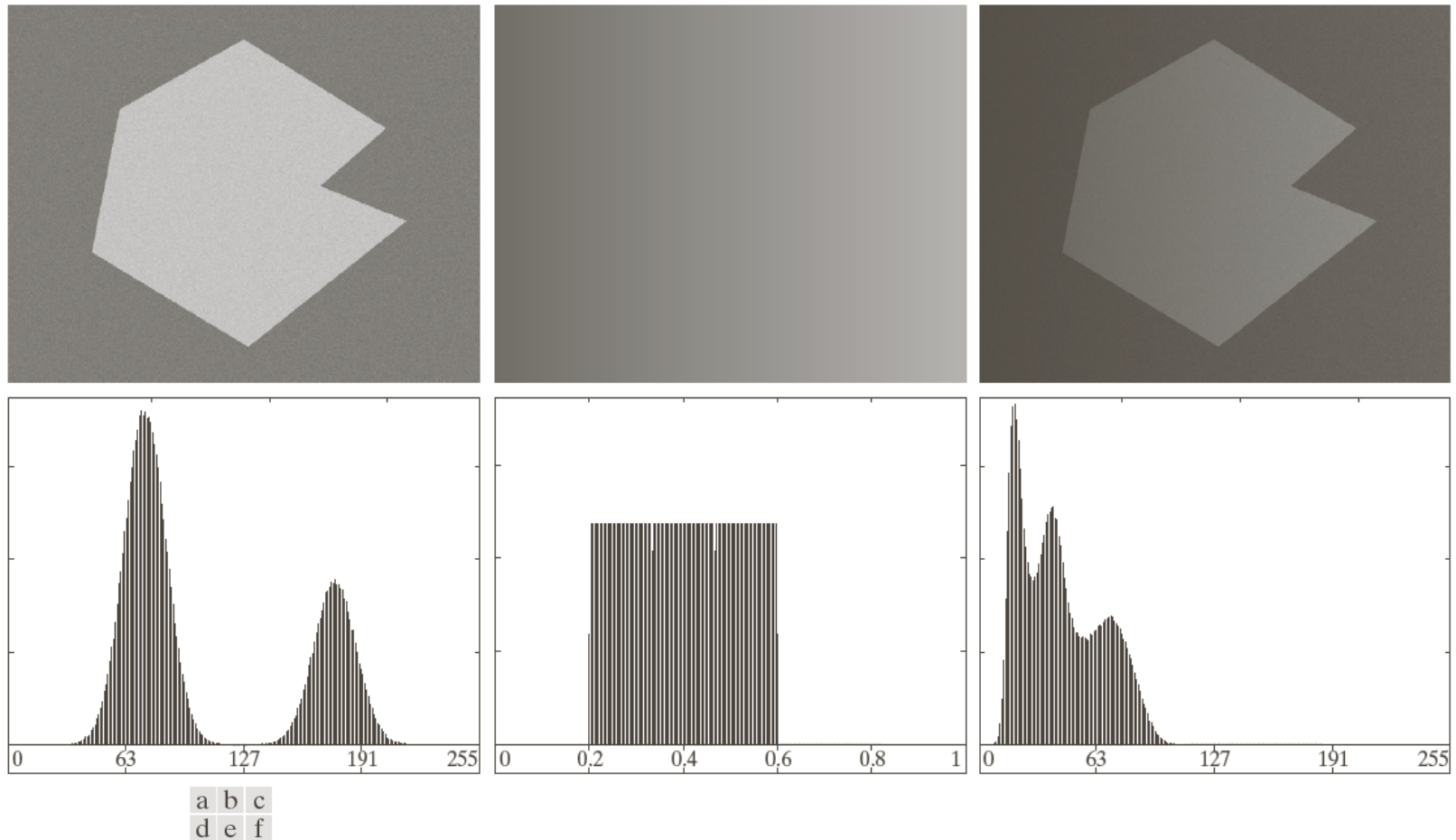
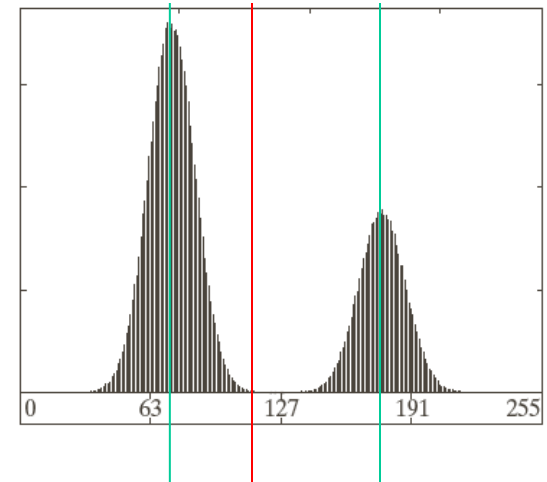
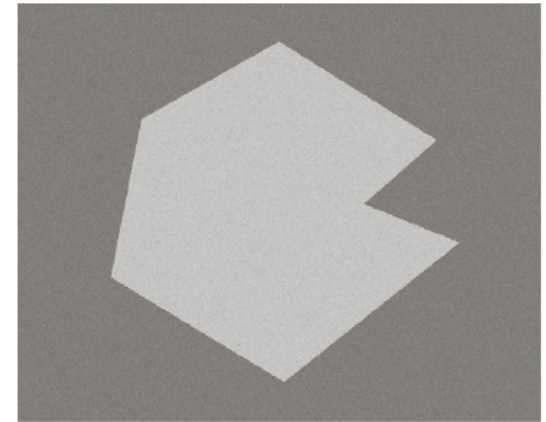


FIGURE 10.37 (a) Noisy image. (b) Intensity ramp in the range $[0.2, 0.6]$. (c) Product of (a) and (b). (d)–(f) Corresponding histograms.

How to Pick the Threshold

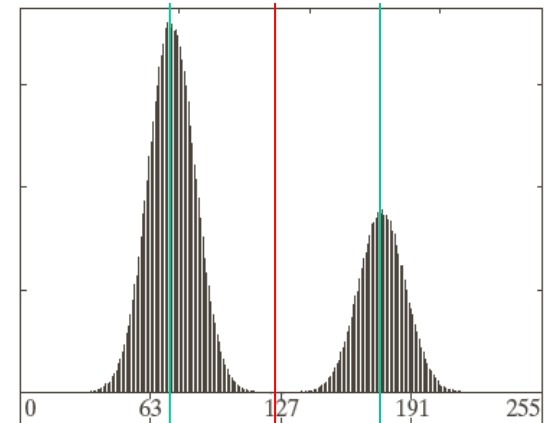
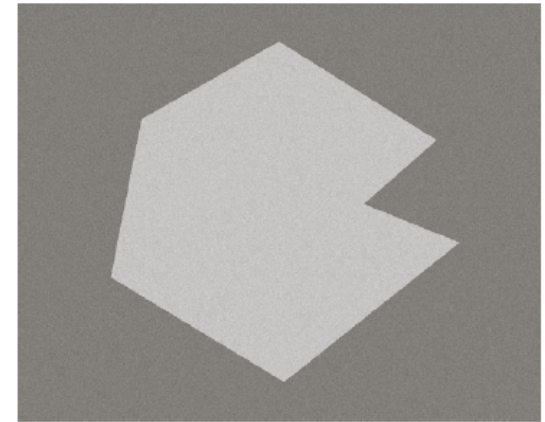
1. Select an initial estimate for the global threshold, T .
2. Segment the image using T by producing two groups of pixels
3. Compute the mean of these two groups of pixels, say m_1 and m_2 .
4. Update the threshold $T=(m_1 + m_2)/2$
5. Repeat Steps 2 through 4 until convergence



m_1 T m_2

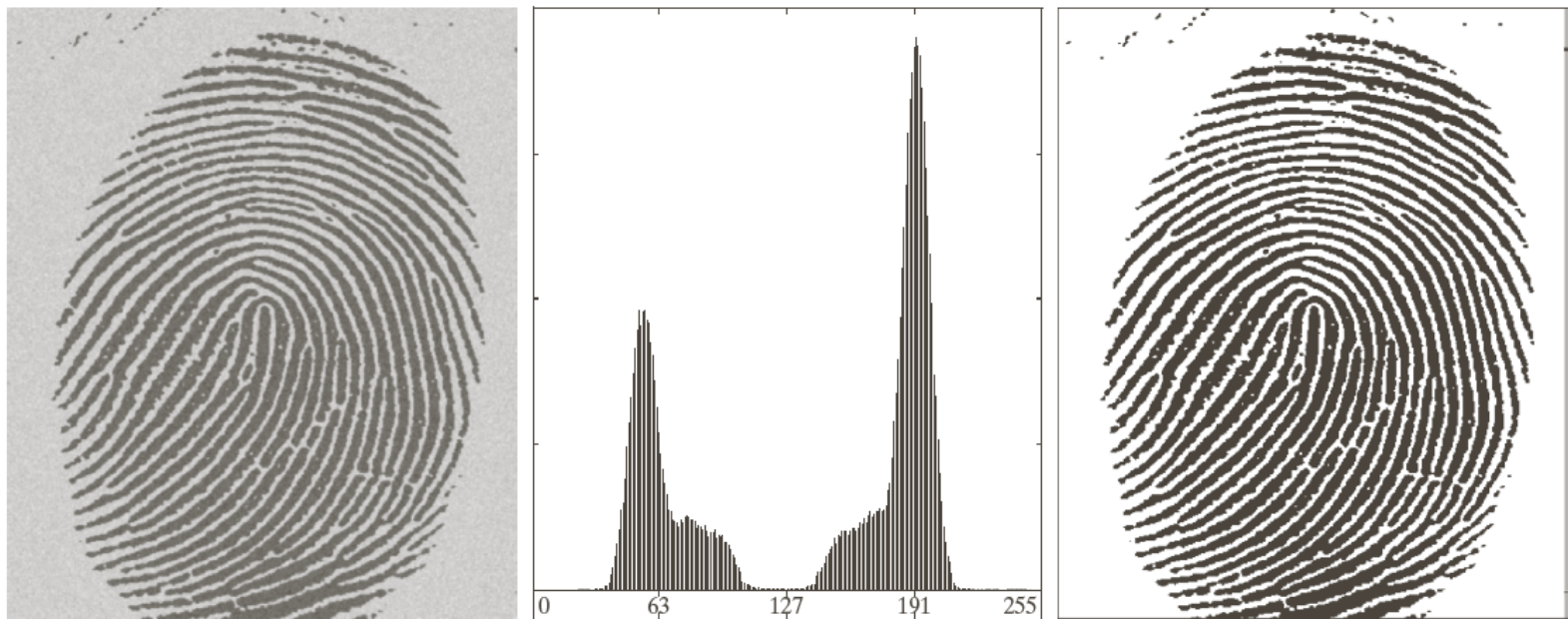
How to Pick the Threshold

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5. Repeat Steps 2 through 4 until convergence



m_1 T m_2

An Example



a b c

FIGURE 10.38 (a) Noisy fingerprint. (b) Histogram. (c) Segmented result using a global threshold (the border was added for clarity). (Original courtesy of the National Institute of Standards and Technology.)

Image Segmentation

$$(a) \bigcup_{i=1}^n R_i = R$$

(b) R_i is a connected set, $i = 1, \dots, n$

$$(c) R_i \cap R_j = \phi, \forall i \neq j$$

$$(d) Q(R_i) = TRUE$$

(e) $Q(R_i \cup R_j) = FALSE$ for adjacent regions R_i and R_j

Two categories based on intensity properties:

- Discontinuity – edge-based algorithms
- Similarity – region-based algorithms

Region-Based Segmentation

- **Region growing**
- **Region splitting and merging**

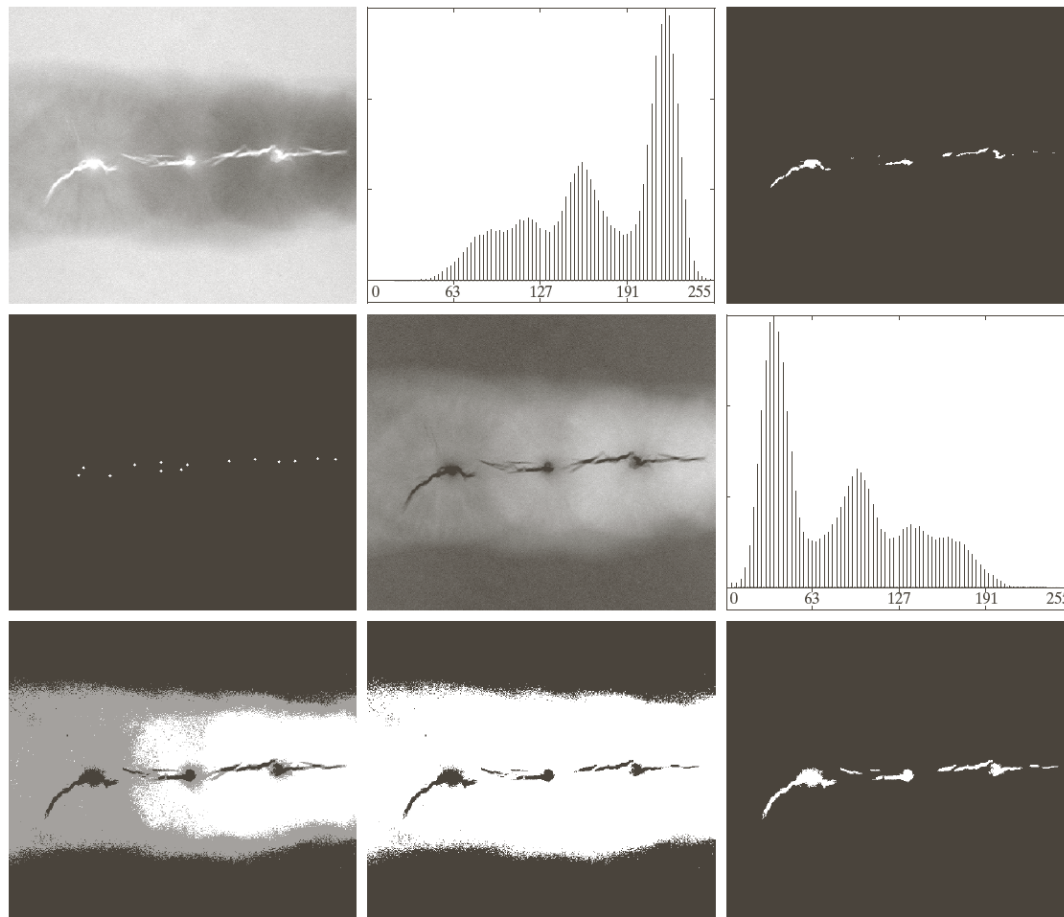
Region Growing Algorithm

-A procedure that groups pixels or subregions into larger regions based on predefined criteria for growth

-Start with a set of “seed” points and grow regions by appending neighboring pixels that satisfy the given criteria

- Connectivity**
- Stopping rules**
 - Local criteria: intensity values, textures, color
 - Prior knowledge: size and shape of the object

An Example



$$Q = \begin{cases} True & \text{if } |I_{seed} - I(x, y)| \leq T \\ False & \text{otherwise} \end{cases}$$

FIGURE 10.51 (a) X-ray image of a defective weld. (b) Histogram. (c) Initial seed image. (d) Final seed image (the points were enlarged for clarity). (e) Absolute value of the difference between (a) and (c). (f) Histogram of (e). (g) Difference image thresholded using dual thresholds. (h) Difference image thresholded with the smallest of the dual thresholds. (i) Segmentation result obtained by region growing. (Original image courtesy of X-TEK Systems, Ltd.)

Region-Based Segmentation

- Region growing
- **Region splitting and merging**

Region-Splitting and Merging Algorithm

Step1: Keep splitting the region while $Q(R_i) = FALSE$ and $R_i > \min Size$

Step 2: Merge the subregions while $Q(R_i \cup R_j) = TRUE$

Region-Splitting and Merging Algorithm

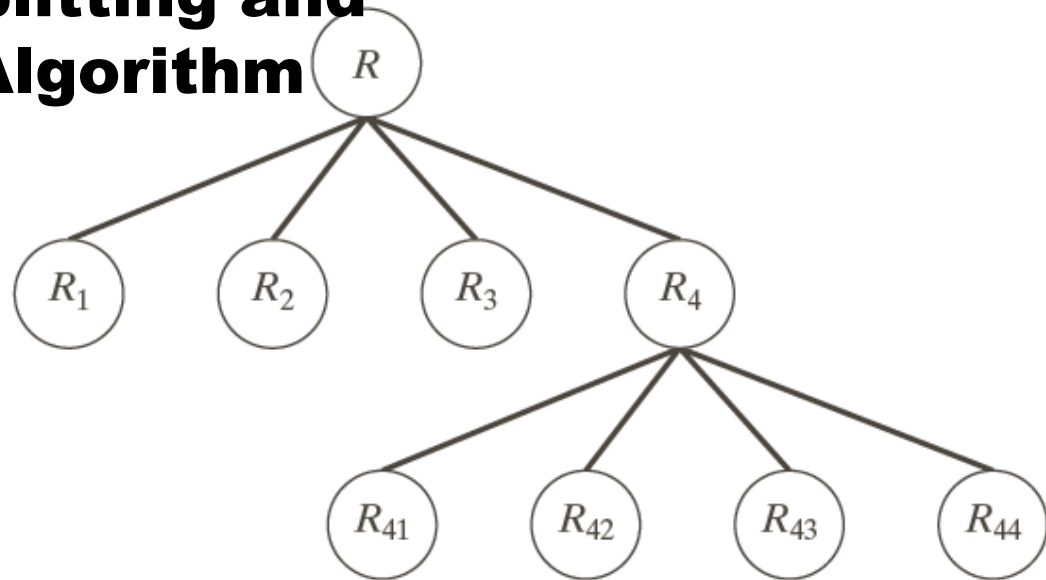
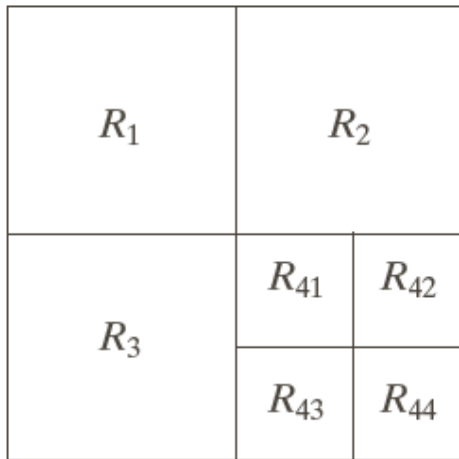
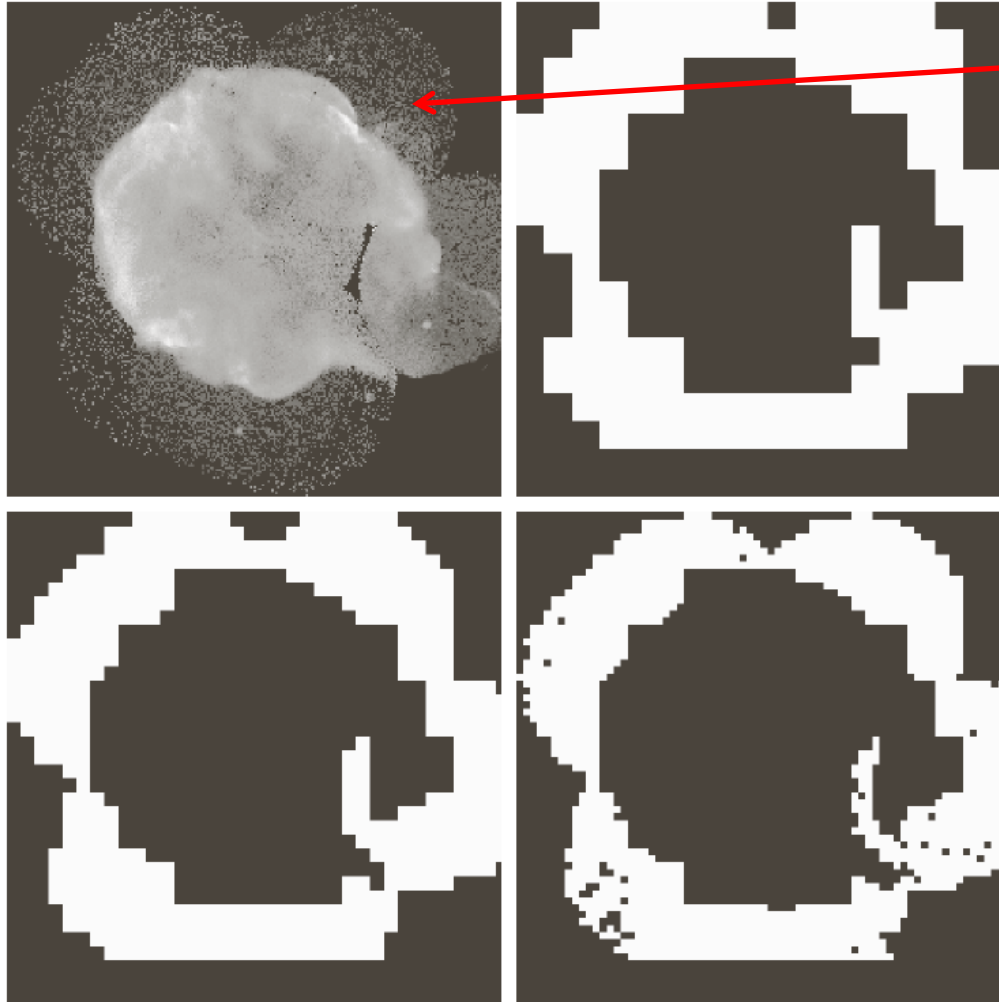


FIGURE 10.52
(a) Partitioned image.
(b) Corresponding quadtree. R represents the entire image region.

An Example



a	b
c	d

FIGURE 10.53

(a) Image of the Cygnus Loop supernova, taken in the X-ray band by NASA's Hubble Telescope. (b)–(d) Results of limiting the smallest allowed quadregion to sizes of 32×32 , 16×16 , and 8×8 pixels, respectively. (Original image courtesy of NASA.)

Extract the outer ring

$$Q = \begin{cases} TRUE & \text{if } \sigma > a \text{ \& } 0 < m < b \\ FALSE & \text{otherwise} \end{cases}$$

Advanced Approaches for Image Segmentation

Image segmentation is still a research problem that is being investigated by many researchers

General-purpose image segmentation is far from well solved

- Image segmentation by K-means clustering
- Image segmentation with Graphic Models (MRF, CRF, etc.)

Semantic Segmentation with Deep Learning

<http://blog.qure.ai/notes/semantic-segmentation-deep-learning-review>